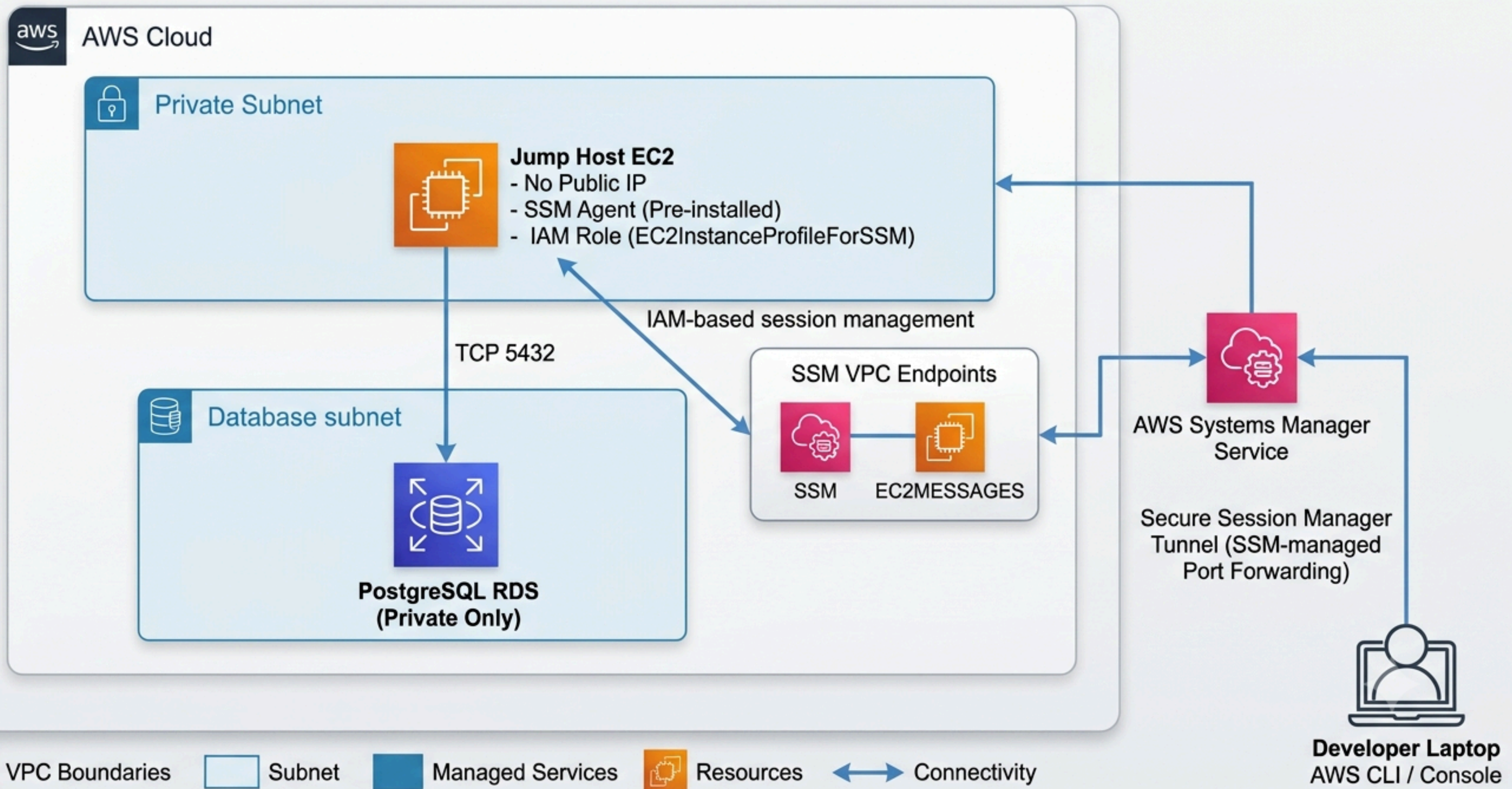




Project

**AWS SSM Jump Host for
Secure Private Resource
Access**

Architecture Diagram



Step 1: Create the VPC

h

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1

Asia Pacific (Mumbai)

Sejaal (389813258309)

Sejaal

003b582c2d3ea8f16

vpc-003b582c2d3ea8f16 / ssm-jump-host-vpc

Actions

Details

Info

VPC ID

vpc-003b582c2d3ea8f16

DNS resolution

Enabled

Main network ACL

acl-0ffdef089db70f511

IPv6 CIDR (Network border group)

—

Encryption control ID

—

State

Available

Tenancy

default

Default VPC

No

Network Address Usage metrics

Disabled

Encryption control mode

—

Block Public Access

Off

DHCP option set

dopt-03ef7f78d77f89406

IPv4 CIDR

10.0.0.0/16

Route 53 Resolver DNS Firewall rule groups

—

DNS hostnames

Enabled

Main route table

rtb-0d80e9ab48ed5941e

IPv6 pool

—

Owner ID

389813258309

Step 2 : Create Private Subnet 1.

aws

Search

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1

Asia Pacific (Mumbai)

Sejaal

VPC

Subnets

subnet-0683d8727e3fb57bd

subnet-0683d8727e3fb57bd / ssm-jump-host-private-subnet-1

Details

Subnet ID

subnet-0683d8727e3fb57bd

IPv4 CIDR

10.0.1.0/24

Availability Zone

aps1-az1 (ap-south-1a)

Network ACL

-

Auto-assign customer-owned IPv4 address

No

IPv6 CIDR reservations

-

Resource name DNS AAAA record

Disabled

Subnet ARN

arn:aws:ec2:ap-south-1:389813258309:subnet/subnet-0683d8727e3fb57bd

Available IPv4 addresses

251

Network border group

ap-south-1

Default subnet

No

Customer-owned IPv4 pool

-

IPv6-only

No

DNS64

Disabled

State

Available

IPv6 CIDR

-

VPC

vpc-003b582c2d3ea8f16 | ssm-jump-host-vpc

Auto-assign public IPv4 address

No

Outpost ID

-

Hostname type

IP name

Owner

389813258309

Block Public Access

Off

IPv6 CIDR association ID

-

Route table

-

Auto-assign IPv6 address

No

IPv4 CIDR reservations

-

Resource name DNS A record

Disabled

Step 3 : Create Private Subnet 2

aws

Search

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Ask Amazon Q

1

Asia Pacific (Mumbai)

Sejaal

VPC

Subnets

subnet-037a3be6d1a9ef1b7

subnet-037a3be6d1a9ef1b7 / ssm-jump-host-private-subnet-2

Details

Subnet ID

subnet-037a3be6d1a9ef1b7

IPv4 CIDR

10.0.2.0/24

Availability Zone

aps1-az3 (ap-south-1b)

Network ACL

acl-0ffdef089db70f511

Auto-assign customer-owned IPv4 address

No

IPv6 CIDR reservations

—

Resource name DNS AAAA record

Disabled

Subnet ARN

arn:aws:ec2:ap-south-1:389813258309:subnet/subnet-037a3be6d1a9ef1b7

Available IPv4 addresses

251

Network border group

ap-south-1

Default subnet

No

Customer-owned IPv4 pool

—

IPv6-only

No

DNS64

Disabled

State

Available

IPv6 CIDR

—

VPC

vpc-003b582c2d3ea8f16 | ssm-jump-host-vpc

Auto-assign public IPv4 address

No

Outpost ID

—

Hostname type

IP name

Owner

389813258309

Block Public Access

Off

IPv6 CIDR association ID

—

Route table

rtb-0d80e9ab48ed5941e

Auto-assign IPv6 address

No

IPv4 CIDR reservations

—

Resource name DNS A record

Disabled

Step 4 : Create a Private Route Table

aws

Search

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Ask Amazon Q

1

Asia Pacific (Mumbai)

Sejaal (389813258309)

Sejaal

VPC

Route tables

rtb-034c1630035dbd715

rtb-034c1630035dbd715 / ssm-jump-host-private-rt

Actions

Details

Info

Route table ID

rtb-034c1630035dbd715

VPC

vpc-003b582c2d3ea8f16 | ssm-jump-host-vpc

Main

No

Owner ID

389813258309

Explicit subnet associations

—

Edge associations

—

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

Routes (1)

Both

Edit routes

Filter routes

< 1 >

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	Create Route Table

Step 4.2 : Associate Both Private Subnets

aws

Search

[Alt+S] Ask Amazon Q

1

Asia Pacific (Mumbai)

Sejaal (389813258309)

VPC

Route tables

rtb-034c1630035dbd715

rtb-034c1630035dbd715 / ssm-jump-host-private-rt

Actions

Details

Info

Route table ID

rtb-034c1630035dbd715

VPC

vpc-003b582c2d3ea8f16 | ssm-jump-host-vpc

Main

No

Owner ID

389813258309

Explicit subnet associations

2 subnets

Edge associations

—

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

Explicit subnet associations (2)

Edit subnet associations

Find subnet association

< 1 >

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
ssm-jump-host-private-subnet-1	subnet-0683d8727e3fb57bd	10.0.1.0/24	—
ssm-jump-host-private-subnet-2	subnet-037a3be6d1a9ef1b7	10.0.2.0/24	—

Step 5 : Create a Security Group for SSM VPC Endpoints

aws

Search

[Alt+S] Ask Amazon Q

1

Asia Pacific (Mumbai) ▼

☰

VPC > Security Groups > sg-09695ca66eb1e073e

sg-09695ca66eb1e073e - ssm-vpc-endpoint-sg

Details

Security group name ☐ ssm-vpc-endpoint-sg	Security group ID ☐ sg-09695ca66eb1e073e	Description ☐ Allow HTTPS access to SSM VPC endpoints from Jump Host	VPC ID vpc-003b582c2d3ea8f16
Owner ☐ 389813258309	Inbound rules count 0 Permission entries	Outbound rules count 1 Permission entry	

Step 6 : Create SSM VPC Endpoints

aws

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Ask Amazon Q

1

Asia Pacific (Mumbai)

Sejaal (389813258309)

Sejaal

VPC

Endpoints

vpce-00ff2fc41995b2a66

vpce-00ff2fc41995b2a66 / ssm-jump-host-ssm-endpoint

Actions

Details

Endpoint ID vpce-00ff2fc41995b2a66	Status Available	Creation time Tuesday, August 25, 2026 at 19:17:05 GMT+5:30	Endpoint type Interface
VPC ID vpc-003b582c2d3ea8f16 (ssm-jump-host-vpc)	Status message -	Service name com.amazonaws.ap-south-1.ssm	Private DNS names enabled Yes
DNS record IP type ipv4	IP address type ipv4	Service region ap-south-1	DNS names vpce-00ff2fc41995b2a66-ofiop8uw.ssm.ap-south-1.vpce.amazonaws.com-(Z2KVTB3ZLFM7JR) vpce-00ff2fc41995b2a66-ofiop8uw-ap-south-1b.ssm.ap-south-1.vpce.amazonaws.com-(Z2KVTB3ZLFM7JR) vpce-00ff2fc41995b2a66-ofiop8uw-ap-south-1a.ssm.ap-south-1.vpce.amazonaws.com-(Z2KVTB3ZLFM7JR) ssm.ap-south-1.amazonaws.com-(Z0413283174ZSX5V1T2LJ) ssm.ap-south-1.api.aws-(Z04133103N1WDXCKEH7M8)
Private DNS names 2	Private DNS only inbound resolver endpoint -	Private DNS preference -	

Step 6.2 : Create the ssmessages VPC Endpoint

aws

Search

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Asia Pacific (Mumbai)

Sejaal (389813258309)

Sejaal

VPC

Endpoints

vpce-0c432919d1d8446f4

vpce-0c432919d1d8446f4 / ssm-jump-host-ssmmessages-endpoint

Actions

Details

Endpoint ID

vpce-0c432919d1d8446f4

VPC ID

vpce-003b582c2d3ea8f16 (ssm-jump-host-vpc)

DNS record IP type

ipv4

Private DNS names

2

Status

Available

Status message

IP address type

ipv4

Private DNS only inbound resolver endpoint

Creation time

Tuesday, August 25, 2026 at 19:27:37 GMT+5:30

Service name

com.amazonaws.ap-south-1.ssmessages

Service region

ap-south-1

Private DNS preference

Endpoint type

Interface

Private DNS names enabled

Yes

DNS names

vpce-0c432919d1d8446f4-4rb72hmt.ssmessages.ap-south-1.vpc.amazonaws.com- (Z2KVTB3ZLFM7JR)

vpce-0c432919d1d8446f4-4rb72hmt-ap-south-1b.ssmessages.ap-south-1.vpc.amazonaws.com- (Z2KVTB3ZLFM7JR)

vpce-0c432919d1d8446f4-4rb72hmt-ap-south-1a.ssmessages.ap-south-1.vpc.amazonaws.com- (Z2KVTB3ZLFM7JR)

ssmmessages.ap-south-1.amazonaws.com- (Z04344963SDLQSI0GUY6A)

ssmmessages.ap-south-1.api.aws- (Z04222893C97QRKUEDNZL)

Next: Step 6.3 : Create the ec2messages VPC Endpoint

VPC > Endpoints > vpce-04f0bfdebcc2d32d4

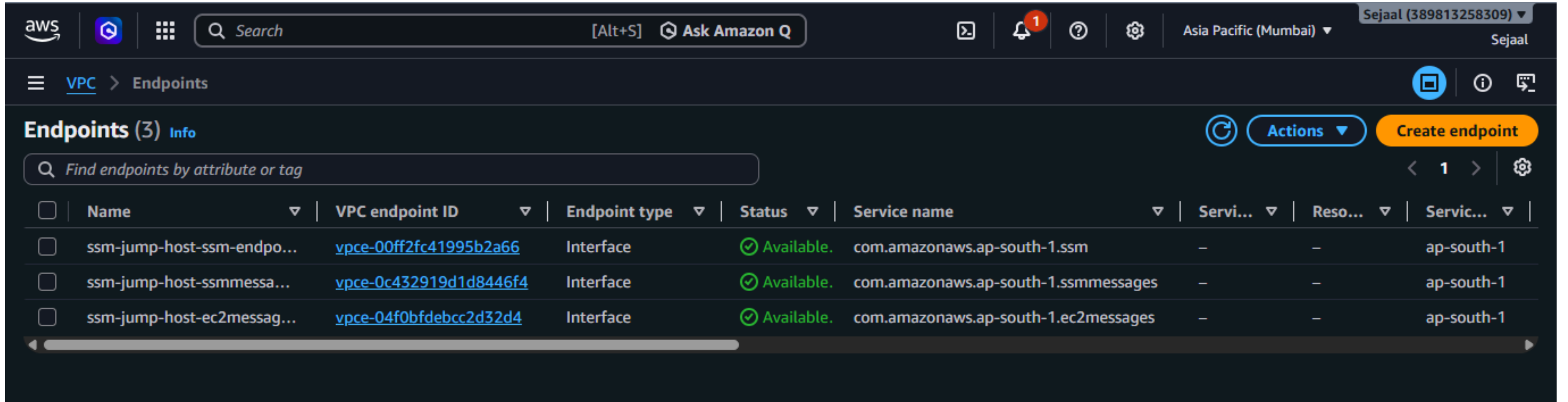
vpce-04f0bfdebcc2d32d4 / ssm-jump-host-ec2messages-endpoint

Actions

Details

Endpoint ID vpce-04f0bfdebcc2d32d4	Status Available	Creation time Tuesday, August 25, 2026 at 19:32:40 GMT+5:30	Endpoint type Interface
VPC ID vpc-003b582c2d3ea8f16 (ssm-jump-host-vpc)	Status message —	Service name com.amazonaws.ap-south-1.ec2messages	Private DNS names enabled Yes
DNS record IP type ipv4	IP address type ipv4	Service region ap-south-1	DNS names vpce-04f0bfdebcc2d32d4-a7nyze5c.ec2messages.ap-south-1.vpce.amazonaws.com- (Z2KVTB3ZLFM7JR) vpce-04f0bfdebcc2d32d4-a7nyze5c-ap-south-1b.ec2messages.ap-south-1.vpce.amazonaws.com- (Z2KVTB3ZLFM7JR) vpce-04f0bfdebcc2d32d4-a7nyze5c-ap-south-1a.ec2messages.ap-south-1.vpce.amazonaws.com- (Z2KVTB3ZLFM7JR) ec2messages.ap-south-1.amazonaws.com- (Z0273275287JQBEFDZUFY)
Private DNS names ec2messages.ap-south-1.amazonaws.com	Private DNS only inbound resolver endpoint —	Private DNS preference —	

All three endpoints



The screenshot shows the AWS Management Console interface for VPC Endpoints. At the top, there's a navigation bar with the AWS logo, a search bar, and a user profile for 'Sejaal (389813258309)'. Below the navigation bar, the breadcrumb trail shows 'VPC > Endpoints'. The main heading is 'Endpoints (3) Info'. There's a search bar with the placeholder text 'Find endpoints by attribute or tag'. To the right of the search bar are buttons for 'Actions' and 'Create endpoint'. Below these is a table with three endpoints. The table has columns for Name, VPC endpoint ID, Endpoint type, Status, Service name, and several other columns. All three endpoints are of type 'Interface' and have a status of 'Available'.

☐	Name	VPC endpoint ID	Endpoint type	Status	Service name	Servi...	Reso...	Servic...
☐	ssm-jump-host-ssm-endpo...	vpce-00ff2fc41995b2a66	Interface	✓ Available.	com.amazonaws.ap-south-1.ssm	–	–	ap-south-1
☐	ssm-jump-host-ssmmessa...	vpce-0c432919d1d8446f4	Interface	✓ Available.	com.amazonaws.ap-south-1.ssmmessages	–	–	ap-south-1
☐	ssm-jump-host-ec2messag...	vpce-04f0bfdebcc2d32d4	Interface	✓ Available.	com.amazonaws.ap-south-1.ec2messages	–	–	ap-south-1

Note: All 3 endpoints are connected to the ssm-vpc-endpoint-sg Security Group

Step 7 : Create the IAM Role for the Jump Host

Role details

ssm-jump-host-role [Info](#)

IAM role for private EC2 Jump Host managed through AWS Systems Manager

Summary

Creation date

August 25, 2026, 19:59 (UTC+05:30)

ARN

 arn:aws:iam::389813258309:role/ssm-jump-host-role

Instance profile ARN

 arn:aws:iam::389813258309:instance-profile/ssm-jump-host-role

Last activity

-

Maximum session duration

1 hour

Delete

Edit

Permissions tab showing the policy

Permissions

Trust relationships

Tags

Last Accessed

Revoke sessions

Permissions policies (1) [Info](#)

You can attach up to 20 managed policies.

 [Simulate](#) [Remove](#) [Add permissions](#)

Search

Filter by Type

All types

< 1 >



☐	Policy name ↗	Type	Attached entities
☐	  AmazonSSMManagedInstanceCore	AWS managed	1

Step 8 : Create the Jump Host Security Group

Now we need to allow the Jump Host to reach the three SSM VPC endpoints over HTTPS.

Go to: VPC → Security Groups → ssm-vpc-endpoint-sg

Then: Open Inbound rules → Click Edit inbound rules → Click Add rule

Configure: Type → HTTPS -> Protocol → TCP → Port range → 443 → Source -> Custom → ssm-jump-host-sg

VPC

Security Groups

sg-09ecbd57c23e22179

sg-09ecbd57c23e22179

ssm-jump-host-sg

Details

Security group name

ssm-jump-host-sg

Security group ID

sg-09ecbd57c23e22179

Description

Security group for private SSM managed J
ump Host

VPC ID

[vpc-003b582c2d3ea8f16](#)

Owner

389813258309

Inbound rules count

0 Permission entries

Outbound rules count

1 Permission entry

Step 8.2 : Configure the SSM Endpoint Security Group

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (1)

Q Search

Manage tags

Edit inbound rules

< 1 >

⚙

☐	Name ▾	Security group rule ID ▾	IP version ▾	Type ▾	Protocol ▾	Port ra... ▾	Source ▾	Description
☐	-	sgr-08e1379b40487cd67	-	HTTPS	TCP	443	sg-09ecbd57c23e22179 / ssm-jump-host-sg	-

Step 9 : Launch the Private EC2 Jump Host

aws

Search

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1

Asia Pacific (Mumbai)

Sejaal (38981325830)

EC2

Instances

i-0e24c09235ec9db01

Instance summary for i-0e24c09235ec9db01 (ssm-jump-host) Info

Updated 3 minutes ago

Instance ID

i-0e24c09235ec9db01

IPv6 address

–

Hostname type

IP name: ip-10-0-1-112.ap-south-1.compute.internal

Answer private resource DNS name

–

Auto-assigned IP address

–

IAM role

ssm-jump-host-role

IMDSv2

Required

Public IPv4 address

–

Instance state

Running

Private IP DNS name (IPv4 only)

ip-10-0-1-112.ap-south-1.compute.internal

Instance type

t3.micro

VPC ID

vpc-003b582c2d3ea8f16 (ssm-jump-host-vpc)

Subnet ID

subnet-0683d8727e3fb57bd (ssm-jump-host-private-subnet-1)

Instance ARN

arn:aws:ec2:ap-south-1:389813258309:instance/i-0e24c09235ec9db01

Private IPv4 addresses

10.0.1.112

Public DNS

–

Elastic IP addresses

–

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name

–

Managed

false

Connect

Instance state

Actions

aws

Search

[Alt+S]

1

Asia Pacific (Mumbai)

Sejaal (389813258309)

Sejaal

EC2

Instances

i-0e24c09235ec9db01

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

VPC ID

vpc-003b582c2d3ea8f16 (ssm-jump-host-vpc)

Subnet ID

subnet-0683d8727e3fb57bd (ssm-jump-host-private-subnet-1)

Availability zone

ap-south-1a

Availability zone ID

aps1-az1

Outpost ID

—

IP addresses

Info

Public IPv4 address

—

Private IPv4 addresses

10.0.1.112

IPv6 addresses

—

Secondary private IPv4 addresses

—

Carrier IP addresses (ephemeral)

—

Hostname and DNS

Info

Public DNS

—

Private IP DNS name (IPv4 only)

ip-10-0-1-112.ap-south-1.compute.internal

IPv4-only IP based name: A record only

—

Dualstack - IP based name: A and AAAA record

—

IPv6-only - IP based name: AAAA record only

—

Public hostname type

—

Private hostname type

IP name: ip-10-0-1-112.ap-south-1.compute.internal

Use RBN as guest OS hostname

Disabled

Answer RBN DNS hostname IPv4

Disabled

EC2 networking/details showing:

aws

Search

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1

Asia Pacific (Mumbai)

Sejaal (389813258)

EC2

Instances

i-0e24c09235ec9db01

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Security details

IAM role

ssm-jump-host-role

Owner ID

389813258309

Launch time

Wed Aug 26 2026 19:04:09 GMT+0530 (India Standard Time)

Security groups

sg-09ecbd57c23e22179 (ssm-jump-host-sg)

▼ Inbound rules

Filter rules

< 1

Name	Security group rule ID	Port range	Protocol	Source	Security groups
No rules to display					

▼ Outbound rules

Filter rules

< 1

IAM/security details showing

Step 10 : Verify the Jump Host in AWS Systems Manager

AWS Console → Systems Manager → Fleet Manager → Managed nodes

Look for: i-0e24c09235ec9db01; or the instance named: ssm-jump-host

What we want to see: Ping status: Online

If it becomes Online, it proves that:

the IAM role is working

the SSM Agent is working

the private VPC endpoints are working

no NAT Gateway is required

no public IP is required

Managed Nodes (1)									
Filter									
Last fetched at: 7:30 PM									
☐	Node ID	Node s...	Name	Platfor...	Operatin...	Resourc...	Source ID	Ping st...	Agent ...
☐	i-0e24c09235ec9db01	Running	ssm-jump-host	Linux	Amazon Linux	EC2 instance	-	Online	3.3.4624

Step 11 : Create the Security Group for PostgreSQL RDS

Before creating the database, we first create a dedicated Security Group.

Why?:- The PostgreSQL RDS instance must be private and its Security Group should only allow traffic from the Jump Host Security Group.

The screenshot displays the AWS Management Console interface for a Security Group. The breadcrumb navigation at the top shows 'VPC > Security Groups > sg-058969ed943c6e1a1'. The main heading is 'sg-058969ed943c6e1a1 - ssm-jump-host-rds-sg'. An 'Actions' button is located in the top right corner.

Details

Security group name ssm-jump-host-rds-sg	Security group ID sg-058969ed943c6e1a1	Description Allow PostgreSQL access only from SSM Jump Host	VPC ID vpc-003b582c2d3ea8f16
Owner 389813258309	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry	

Navigation tabs: **Inbound rules** (selected), Outbound rules, Sharing, VPC associations, Related resources - new, Tags.

Inbound rules (1/1)

Search:

Buttons: Refresh, Manage tags, Edit inbound rules

Page navigation: < 1 > Settings

☒	Name ▾	Security group rule ID ▾	IP version ▾	Type ▾	Protocol ▾	Port range ▾	Source ▾	Description
☒	-	sgr-05efe7db74ee5b803	-	PostgreSQL	TCP	5432	sg-09ecbd57c23e22179 / ssm-jump-host-sg	-

Step 12 : Create an RDS DB Subnet Group

Before launching PostgreSQL RDS, we need to tell RDS which private subnets it is allowed to use. We will create a DB Subnet Group using both of your private subnets This ensures RDS is deployed privately and has subnet availability across two Availability Zones.

AWS Console → RDS → Subnet groups → Create DB subnet group

ssm-jump-host-db-subnet-group

Subnet group details

VPC ID

vpc-003b582c2d3ea8f16

ARN

arn:aws:rds:ap-south-1:389813258309:subgrp:ssm-jump-host-db-subnet-group

Supported network types

IPv4

Description

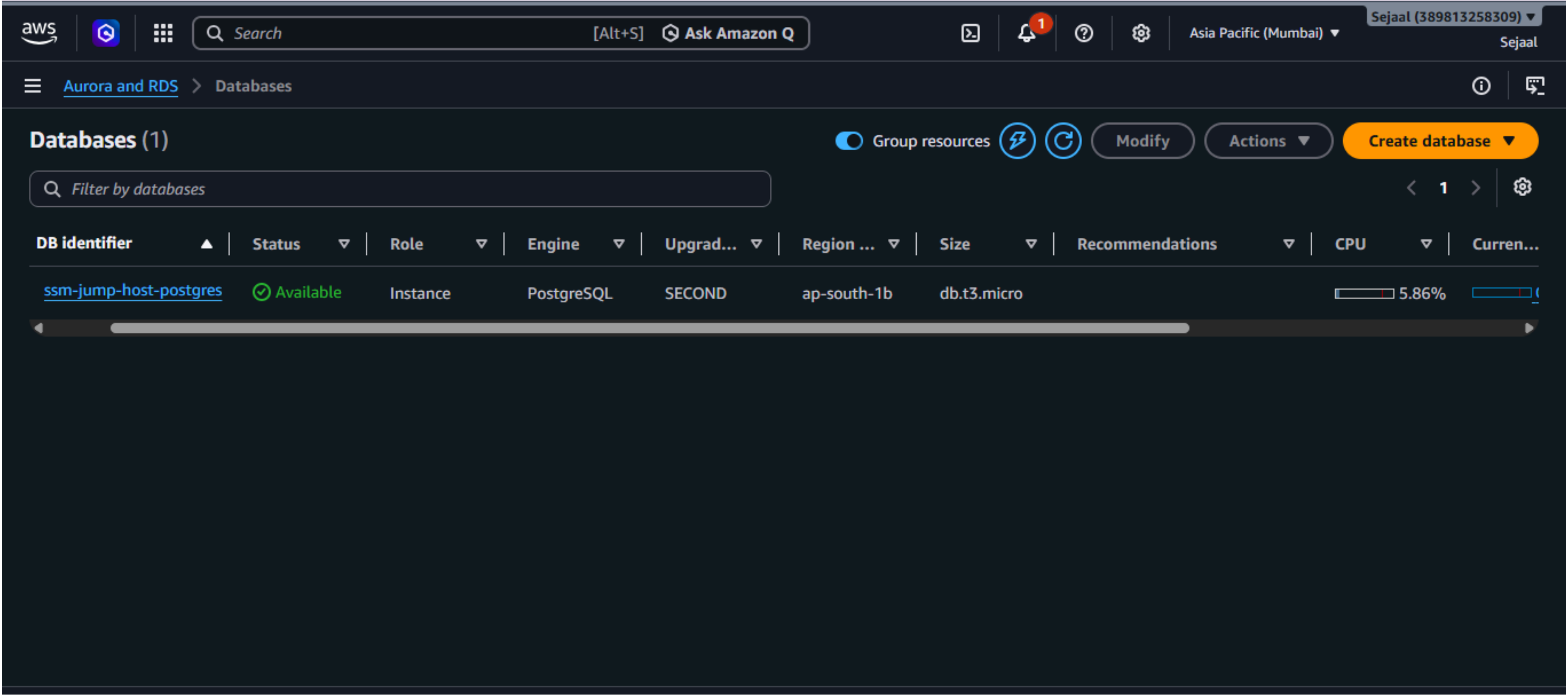
Private subnets for SSM Jump Host PostgreSQL RDS

Subnets (2)

Availability zone	Subnet name	Subnet ID	CIDR block
ap-south-1a	ssm-jump-host-private-subnet-1	subnet-0683d8727e3fb57bd	10.0.1.0/24
ap-south-1b	ssm-jump-host-private-subnet-2	subnet-037a3be6d1a9ef1b7	10.0.2.0/24

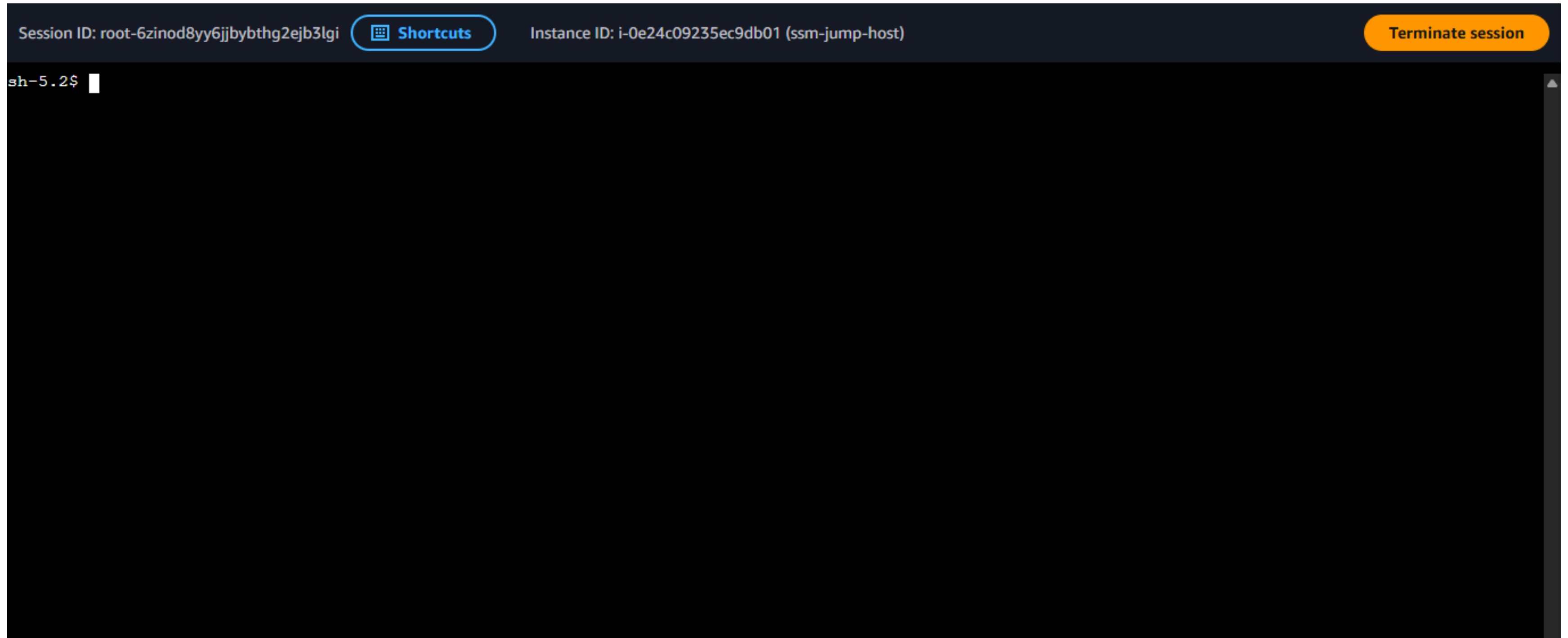
Step 13 : Create the Private PostgreSQL RDS Instance

Now we will create the actual database. AWS Console → RDS → Databases → Create database



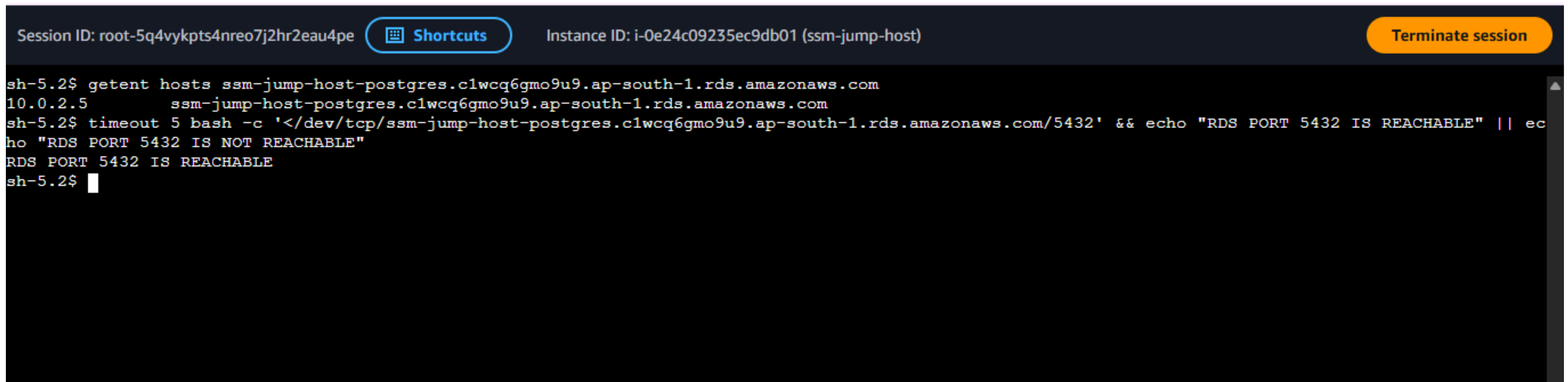
Step 15 : Test SSM Session Manager Connection

AWS Console → Systems Manager → Fleet Manager / Managed nodes → Node actions → Start terminal session



Step 16 — Test Jump Host → Private RDS Connectivity

RDS → Databases → ssm-jump-host-postgres -> Connectivity & Security -> Endpoints → (Copy the endpoint) → come to ssm terminal → execute command → getent hosts ssm-jump-host-postgres.c1wcq6gmo9u9.ap-south-1.rds.amazonaws.com (getent hosts endpointkey)

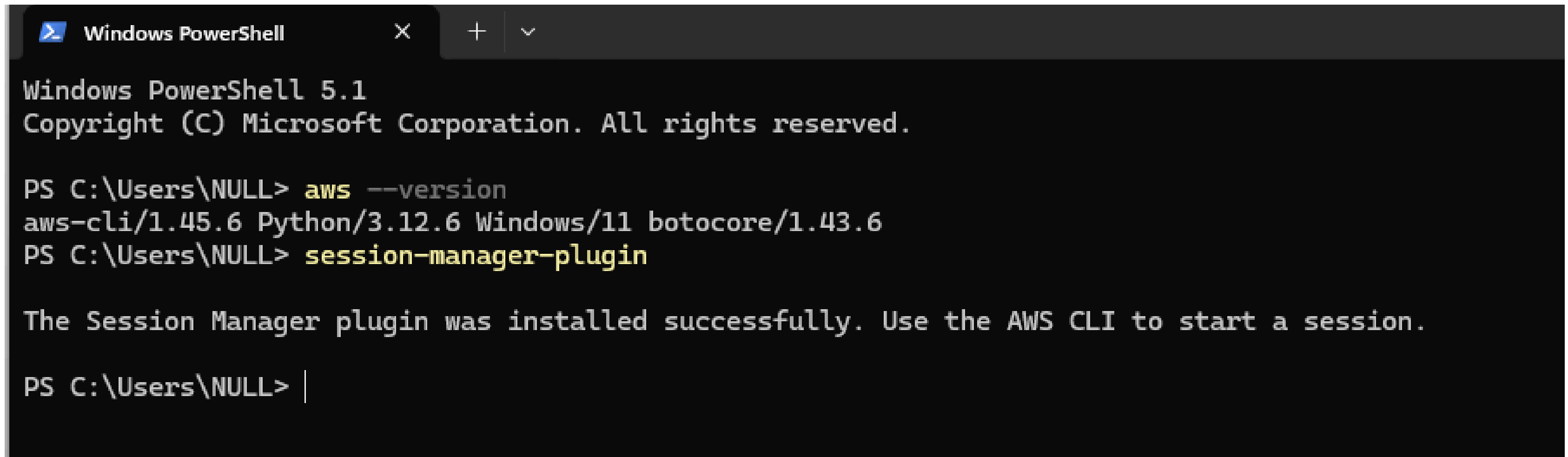


```
Session ID: root-5q4vykpts4nreo7j2hr2eau4pe Shortcuts Instance ID: i-0e24c09235ec9db01 (ssm-jump-host) Terminate session

sh-5.2$ getent hosts ssm-jump-host-postgres.c1wcq6gmo9u9.ap-south-1.rds.amazonaws.com
10.0.2.5      ssm-jump-host-postgres.c1wcq6gmo9u9.ap-south-1.rds.amazonaws.com
sh-5.2$ timeout 5 bash -c '</dev/tcp/ssm-jump-host-postgres.c1wcq6gmo9u9.ap-south-1.rds.amazonaws.com/5432' && echo "RDS PORT 5432 IS REACHABLE" || echo "RDS PORT 5432 IS NOT REACHABLE"
RDS PORT 5432 IS REACHABLE
sh-5.2$
```

Step 17 : Set Up SSM Port Forwarding from Your Local Machine

Open your local terminal PowerShell or Windows Terminal → verify whether the AWS CLI is installed use command → `aws --version` → Then check the Session Manager plugin use command → `session-manager-plugin` → if both are not installed then install it



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\NULL> aws --version
aws-cli/1.45.6 Python/3.12.6 Windows/11 botocore/1.43.6
PS C:\Users\NULL> session-manager-plugin

The Session Manager plugin was installed successfully. Use the AWS CLI to start a session.

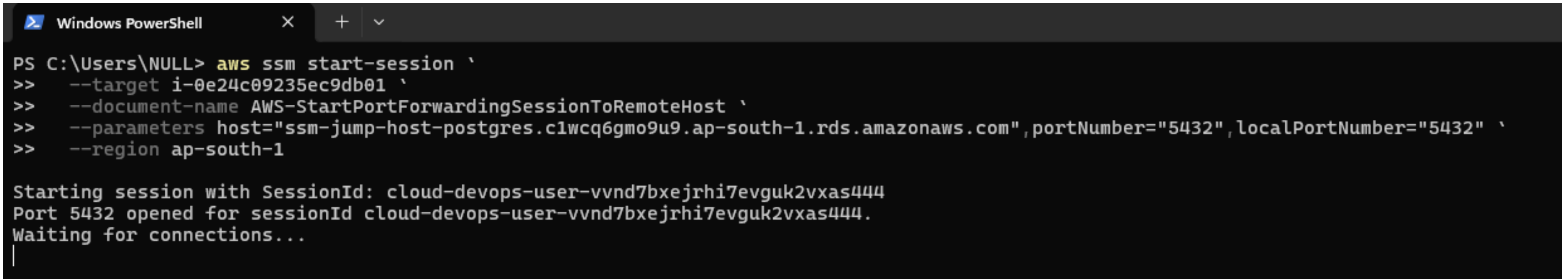
PS C:\Users\NULL> |
```

Checked the availability of aws version and session manager plugin

Step 17.2 : Create the SSM Port Forwarding Tunnel

Run this command in PowerShell:

```
aws ssm start-session `
  --target i-0e24c09235ec9db01 `
  --document-name AWS-StartPortForwardingSessionToRemoteHost `
  --parameters host="ssm-jump-host-postgres.c1wcq6gmo9u9.ap-south-1.rds.amazonaws.com",portNumber="5432",localPortNumber="5432" `
  --region ap-south-1
```

A screenshot of a Windows PowerShell terminal window. The title bar shows 'Windows PowerShell' with standard window controls. The command prompt shows the user at 'C:\Users\NULL' typing the 'aws ssm start-session' command with its arguments. The output indicates that a session was started with a specific SessionId and that port 5432 is now open for connections.

```
PS C:\Users\NULL> aws ssm start-session `
>> --target i-0e24c09235ec9db01 `
>> --document-name AWS-StartPortForwardingSessionToRemoteHost `
>> --parameters host="ssm-jump-host-postgres.c1wcq6gmo9u9.ap-south-1.rds.amazonaws.com",portNumber="5432",localPortNumber="5432" `
>> --region ap-south-1

Starting session with SessionId: cloud-devops-user-vvnd7bxejrhi7evguk2vxas444
Port 5432 opened for sessionId cloud-devops-user-vvnd7bxejrhi7evguk2vxas444.
Waiting for connections...
|
```

SSM Port Forwarding Tunnel is working!

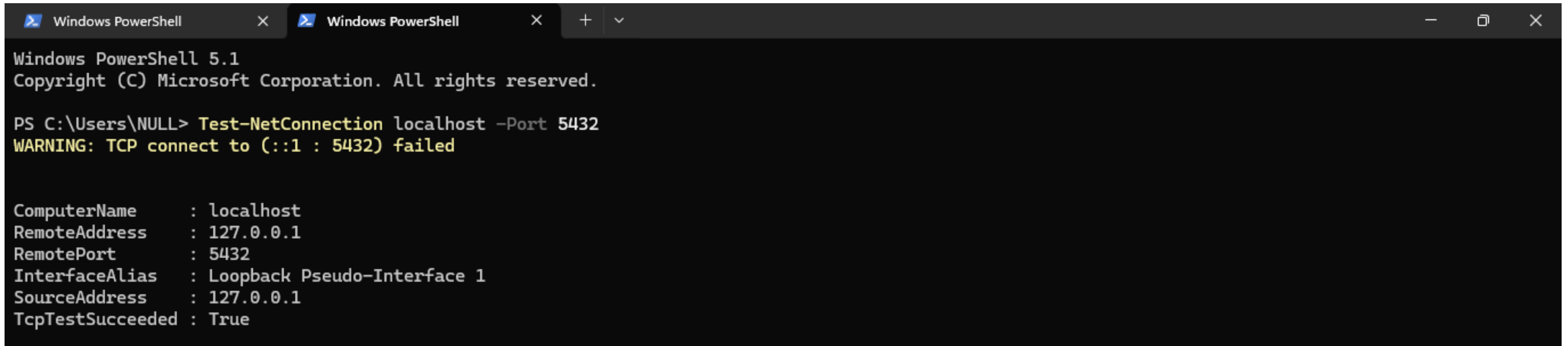
Step 17.3 : Test the Local Tunnel

Keep the current PowerShell window exactly as it is. Do not close it.

Open a second PowerShell window.

Run:

Test-NetConnection localhost -Port 5432

A screenshot of a Windows PowerShell window with a dark background. The window title bar shows two tabs, both labeled 'Windows PowerShell'. The command prompt shows the command 'Test-NetConnection localhost -Port 5432' being executed. The output indicates a successful connection, despite a warning about a failed TCP connect. The output details include the computer name 'localhost', remote address '127.0.0.1', remote port '5432', interface alias 'Loopback Pseudo-Interface 1', source address '127.0.0.1', and a successful TCP test result.

```
Windows PowerShell 5.1
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\NULL> Test-NetConnection localhost -Port 5432
WARNING: TCP connect to (::1 : 5432) failed

ComputerName      : localhost
RemoteAddress     : 127.0.0.1
RemotePort        : 5432
InterfaceAlias    : Loopback Pseudo-Interface 1
SourceAddress     : 127.0.0.1
TcpTestSucceeded  : True
```

local port 5432 is successfully reaching the SSM tunnel.

Step 17.4 : Verify PostgreSQL itself

Keep the SSM port-forwarding tunnel running in PowerShell:

Open DBeaver.

Create/edit a PostgreSQL connection.

Enter: Host, Port, Database, Username, password

Click Test Connection.

Successful result: PostgreSQL 18.3

This confirms: Windows → SSM Tunnel → Jump Host → Private RDS PostgreSQL

```
Windows PowerShell
PS C:\Users\NULL> aws ssm start-session `
>> --target i-0e24c09235ec9db01 `
>> --document-name AWS-StartPortForwardingSessionToRemoteHost `
>> --parameters host="ssm-jump-host-postgres.clwcq6gmo9u9.ap-south-1.rds.amazonaws.com",portNumber="5432",localPortNumber="5432" `
>> --region ap-south-1

Starting session with SessionId: cloud-devops-user-vvnd7bxejrhi7evguk2vxas444
Port 5432 opened for sessionId cloud-devops-user-vvnd7bxejrhi7evguk2vxas444.
Waiting for connections...

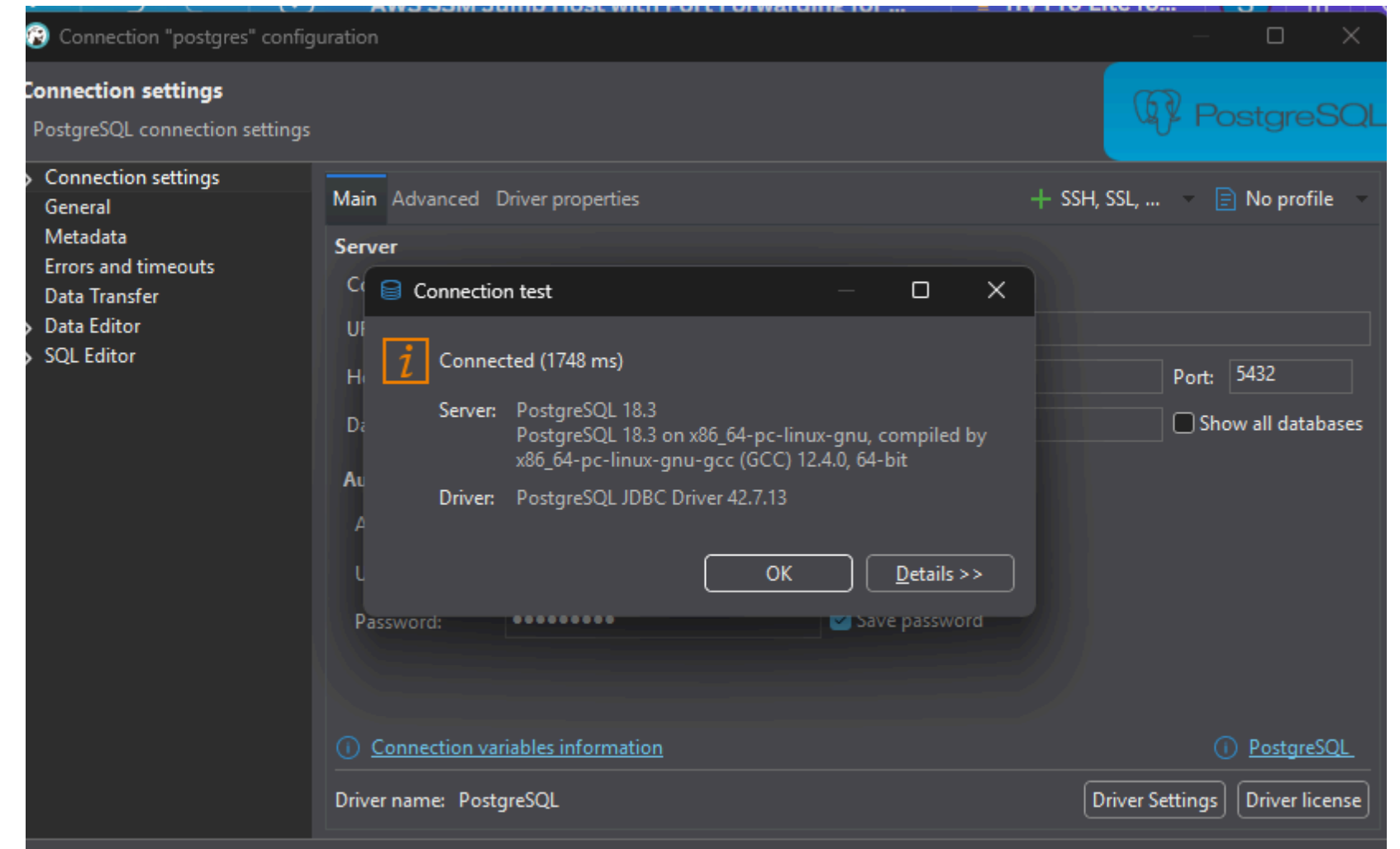
Connection accepted for session [cloud-devops-user-vvnd7bxejrhi7evguk2vxas444]

SessionId: cloud-devops-user-vvnd7bxejrhi7evguk2vxas444 : Your session timed out due to inactivity and has been terminated.

PS C:\Users\NULL> aws ssm start-session `
>> --target i-0e24c09235ec9db01 `
>> --document-name AWS-StartPortForwardingSessionToRemoteHost `
>> --parameters host="ssm-jump-host-postgres.clwcq6gmo9u9.ap-south-1.rds.amazonaws.com",portNumber="5432",localPortNumber="5432" `
>> --region ap-south-1

Starting session with SessionId: cloud-devops-user-8ueedy5prcjsgu2fc4hbt8nzpa
Port 5432 opened for sessionId cloud-devops-user-8ueedy5prcjsgu2fc4hbt8nzpa.
Waiting for connections...

Connection accepted for session [cloud-devops-user-8ueedy5prcjsgu2fc4hbt8nzpa]
```



Step 17.5 : Connect to your private RDS PostgreSQL

We already verified that your Jump Host can reach RDS on port 5432. Now we need to test the actual PostgreSQL login from your Windows machine through the SSM port forwarding.

1. Start the SSM port-forwarding session

Open PowerShell Window 1 and run:

```
aws ssm start-session `
```

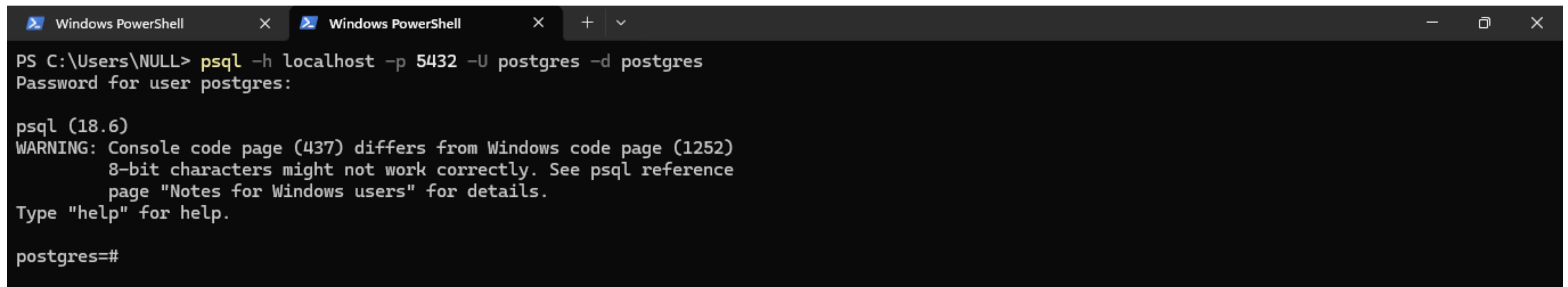
```
--target i-0e24c09235ec9db01 `
```

```
--document-name AWS-StartPortForwardingSessionToRemoteHost `
```

```
--parameters '{"host\":[\"RDS-End-point\"],\"portNumber\":[\"5432\"],\"localPortNumber\":[\"5432\"]}'
```

Then open a second PowerShell window and we'll run:

```
psql -h localhost -p 5432 -U postgres -d postgres
```



```
Windows PowerShell x Windows PowerShell x + v
PS C:\Users\NULL> psql -h localhost -p 5432 -U postgres -d postgres
Password for user postgres:

psql (18.6)
WARNING: Console code page (437) differs from Windows code page (1252)
         8-bit characters might not work correctly. See psql reference
         page "Notes for Windows users" for details.
Type "help" for help.

postgres=#
```

Step 17.6 : Final verification

Since you're already inside psql, run these 3 commands:

```
Windows PowerShell x Windows PowerShell + v
PS C:\Users\NULL> psql -h localhost -p 5432 -U postgres -d postgres
Password for user postgres:

psql (18.6)
WARNING: Console code page (437) differs from Windows code page (1252)
8-bit characters might not work correctly. See psql reference
page "Notes for Windows users" for details.
Type "help" for help.

postgres=# SELECT version();
               version
-----
PostgreSQL 18.6 on x86_64-windows, compiled by msvc-19.44.35228, 64-bit
(1 row)

postgres=# SELECT current_database();
   current_database
-----
postgres
(1 row)

postgres=# SELECT current_user;
   current_user
-----
postgres
(1 row)

postgres=# \q
PS C:\Users\NULL> |
```

Step 18 : Configure HAProxy

```
Windows PowerShell 5.1
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\NULL> Test-NetConnection localhost -Port 15432
WARNING: TCP connect to (::1 : 15432) failed

ComputerName      : localhost
RemoteAddress     : 127.0.0.1
RemotePort        : 15432
InterfaceAlias    : Loopback Pseudo-Interface 1
SourceAddress     : 127.0.0.1
TcpTestSucceeded  : True

PS C:\Users\NULL> notepad C:\haproxy\haproxy.cfg
PS C:\Users\NULL>
```

```
global
    log stdout format raw local0

defaults
    mode tcp
    timeout connect 10s
    timeout client 1m
    timeout server 1m

frontend postgres
    bind *:5432
    default_backend postgres_rds

backend postgres_rds
    server rds 127.0.0.1:15432
```

Ln 16, Col 1 | 254 characte | Plain text | 100% | Windows (C | UTF-8

Step 18 .2 : Verify HAProxy Docker Container

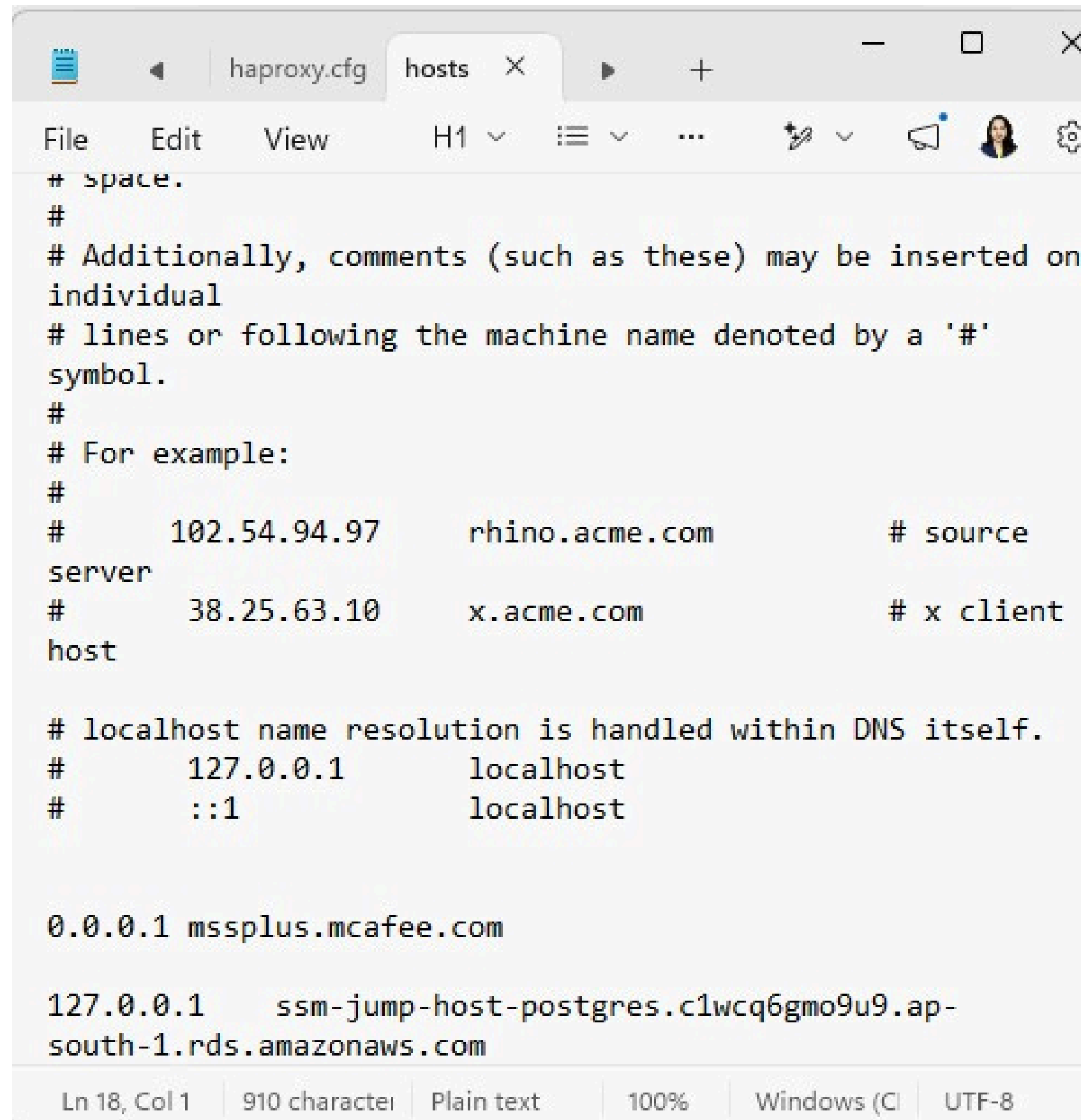
```
Windows PowerShell x Windows PowerShell x + v
PS C:\Users\NULL> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
7210c66016e6   haproxy:3.2   "docker-entrypoint.s..." 38 minutes ago Up 32 minutes 0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
haproxy
PS C:\Users\NULL> |
```

Step 18 .3 : Verify HAProxy Local Port

```
Windows PowerShell x Windows PowerShell x + v
PS C:\Users\NULL> Test-NetConnection localhost -Port 5432

ComputerName      : localhost
RemoteAddress     : ::1
RemotePort        : 5432
InterfaceAlias    : Loopback Pseudo-Interface 1
SourceAddress     : ::1
TcpTestSucceeded  : True
```

hostname mapping



The image shows a code editor window with two tabs: 'haproxy.cfg' and 'hosts'. The 'hosts' tab is active, displaying a configuration file. The editor has a menu bar with 'File', 'Edit', and 'View', and a toolbar with various icons. The status bar at the bottom indicates 'Ln 18, Col 1', '910 character', 'Plain text', '100%', 'Windows (C)', and 'UTF-8'.

```
# space.  
#  
# Additionally, comments (such as these) may be inserted on  
# individual  
# lines or following the machine name denoted by a '#'  
# symbol.  
#  
# For example:  
#  
#      102.54.94.97      rhino.acme.com      # source  
server  
#      38.25.63.10      x.acme.com          # x client  
host  
  
# localhost name resolution is handled within DNS itself.  
#      127.0.0.1        localhost  
#      ::1              localhost  
  
0.0.0.1 mssplus.mcafee.com  
  
127.0.0.1    ssm-jump-host-postgres.c1wcq6gmo9u9.ap-  
south-1.rds.amazonaws.com
```

Step 19 : PostgreSQL End-to-End Connection Test

```
PS C:\Users\NULL> Get-ChildItem "C:\Program Files\PostgreSQL" -Recurse -Filter psql.exe -ErrorAction SilentlyContinue |  
Select-Object -ExpandProperty FullName  
C:\Program Files\PostgreSQL\18\bin\psql.exe  
C:\Program Files\PostgreSQL\18\pgAdmin 4\runtime\psql.exe  
PS C:\Users\NULL> & "C:\Program Files\PostgreSQL\18\bin\psql.exe" -h localhost -p 5432 -U postgres -d postgres  
Password for user postgres:  
  
psql (18.6)  
WARNING: Console code page (437) differs from Windows code page (1252)  
8-bit characters might not work correctly. See psql reference  
page "Notes for Windows users" for details.  
Type "help" for help.  
  
postgres=# SELECT version();  
  
              version  
-----  
PostgreSQL 18.6 on x86_64-windows, compiled by msvc-19.44.35228, 64-bit  
(1 row)  
  
postgres=# |
```

Step 19.1 : Test PostgreSQL using REAL hostname.

```
Windows PowerShell X Windows PowerShell X Windows PowerShell X + v - □ X
PS C:\Users\NULL> & "C:\Program Files\PostgreSQL\18\bin\psql.exe" -h ssm-jump-host-postgres.clwcq6gmo9u9.ap-south-1.rds.amazonaws.com -p 5432 -U postgres
-d postgres
Password for user postgres:

psql (18.6)
WARNING: Console code page (437) differs from Windows code page (1252)
         8-bit characters might not work correctly. See psql reference
         page "Notes for Windows users" for details.
Type "help" for help.

postgres=# \conninfo

```

Parameter	Value
Database	postgres
Client User	postgres
Host	ssm-jump-host-postgres.clwcq6gmo9u9.ap-south-1.rds.amazonaws.com
Host Address	127.0.0.1
Server Port	5432
Options	
Protocol Version	3.0
Password Used	true
GSSAPI Authenticated	false
Backend PID	18940
SSL Connection	false
Superuser	on
Hot Standby	off

```
(13 rows)

postgres=# SELECT version();

```

version
PostgreSQL 18.6 on x86_64-windows, compiled by msvc-19.44.35228, 64-bit

```
(1 row)

postgres=# |
```

Step 20 : Verify TLS

```
postgres"
PS C:\Users\NULL> & "C:\Program Files\PostgreSQL\18\bin\psql.exe" "host=127.0.0.1 port=15432 user=postgres dbname=postgres sslmode=require"
Password for user postgres:

psql (18.6, server 18.3)
WARNING: Console code page (437) differs from Windows code page (1252)
         8-bit characters might not work correctly. See psql reference
         page "Notes for Windows users" for details.
SSL connection (protocol: TLSv1.3, cipher: TLS_AES_256_GCM_SHA384, compression: off, ALPN: postgresql)
Type "help" for help.

postgres=> |
```